

Housing allocation, fertility, and population

Proposed results following the existing household setup

Housing allocation

Borrowing limits can leave housing with old owners even when young households would give up more consumption to use it. In an equilibrium of the model, write $x_i = c_i - \chi n_i$ and $s_i = h_i - \kappa n_i$ for the goods and space available to young adults. The young household's and old owner's marginal values of housing, in current consumption goods, are:

$$MV_i^Y = \frac{\alpha x_i}{s_i}, \quad MV_j^O = \frac{\gamma c_j^2}{h_j^2}. \quad (1)$$

These values measure what each household would pay for a little more space.

The planner reallocates housing and consumption, retaining the physical limits on each tenure and the total housing stock. It may relax private financing restrictions. Each household's fertility, tenure, estate, and future real allocation remain fixed, and existing financial claims are honored.

Proposition 1 (Housing misallocation). *Suppose positive masses of young owners can occupy more housing and old owners can release some. If $MV_i^Y > MV_j^O$ on matched groups, a sufficiently small reallocation toward the young can make them better off and leave every other household equally well off, in the absence of real reallocation costs.*

Proof. Give the young household ϵ units of the old owner's housing. Keeping the old owner's utility fixed requires extra consumption:

$$D_j(\epsilon) = c_j^2 \left[\left(\frac{h_j^2}{h_j^2 - \epsilon} \right)^\gamma - 1 \right]. \quad (2)$$

Take this amount from the young household. Goods and housing balance, and its utility gain has derivative $(MV_i^Y - MV_j^O)/x_i > 0$ at zero. A small change preserves feasibility. Financial settlement replaces the seller's lost estate title with an equal bond claim; the buyer's later sale of the extra title offsets its financing. Future allocations, existing creditors, and aggregate external assets are unchanged. $\square$

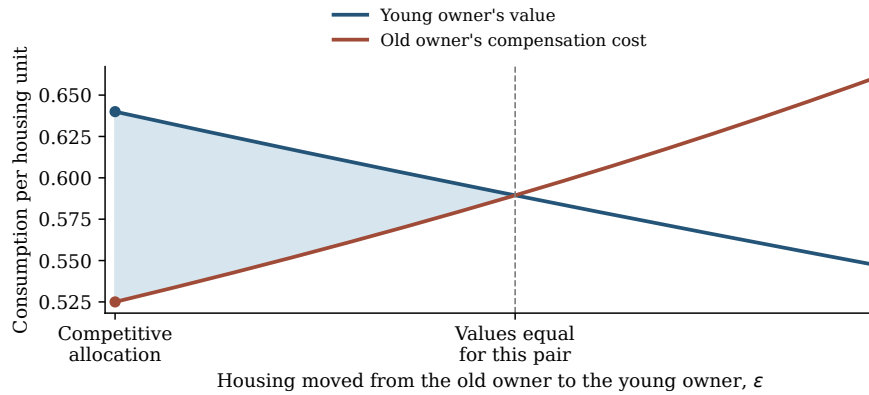


Figure 1. Housing moves toward the young while the old owner is compensated. The shaded gap indicates the direction of improvement; its area is not a welfare measure. Equality of values gives the best allocation along this particular compensated comparison.

Housing access and fertility

The borrowing limit can produce the inequality in Proposition 1. With positive saving, slack physical owner limits, and interior old-age choices, a strictly restrictive down payment gives $MV_i^Y > qr_t$. An old owner with slack retention and estate bounds has $MV_j^O = qr_t$. Young buyers need cash at purchase; old owners already hold their homes.

Rental segmentation limits another way to obtain space. A household may be unable to rent a larger home and unable to fund its purchase. The ownership taste in the model also matters: it can make constrained ownership preferable even when larger rentals are available. Thus both access restrictions can coexist with misallocation, although a binding rental cap is not necessary for the owner-to-owner result. A renter already at a physical cap cannot receive more housing in that tenure under the same planner restrictions.

Real reallocation costs have a simple implication. If moving ϵ housing uses $C(\epsilon)$ goods, with $C(0) = 0$ and $C'(0) = K_{ijt} \geq 0$, the condition becomes $MV_i^Y > MV_j^O + K_{ijt}$. A positive fixed cost instead requires a finite reallocation whose gain covers that cost.

A fertility prediction. More space makes children easier to accommodate; paying for it reduces the goods available to the family. Consider the young owner's conditional problem with positive saving and interior old-age choices. Suppress household and date indices, and let $k = \beta(1 + \gamma + \omega_B)$. The fertility condition and the response of the lifetime budget to more housing are:

$$\frac{\vartheta}{n} = \frac{\chi}{x} + \frac{\alpha\kappa}{s}, \quad (1+k)dx = -\chi dn - p dh. \quad (3)$$

Here p is the net lifetime payment for a marginal unit of current housing, including accompanying transfers. With prices and other resources fixed, $p = qr_t$. Differentiating the first equation using the second gives:

$$\frac{dn}{dh} > 0 \iff \frac{\alpha\kappa}{s^2} > \frac{\chi p}{(1+k)x^2}. \quad (4)$$

Fertility rises when the space benefit exceeds the resource cost. This is a household response while tenure and the binding constraints remain unchanged. Saving and old-age choices are reoptimized. Proposition 1 instead holds future real allocations fixed.

A sufficient condition can be written without prices or a borrowing multiplier. At stationary prices, with zero property tax and a strictly binding down-payment limit, it is:

$$\frac{(1-q)b}{(1-\phi)(y+b)} \geq \frac{\alpha}{1 + \beta(1 + \gamma + \omega_B) + \alpha}. \quad (5)$$

The left side is housing expenditure at the purchase limit relative to lifetime resources. Under the stated household conditions, this restriction ensures a fertility increase for any $p < MV^Y$. The full equilibrium response also depends on prices, tenure choices, and subsequent cohort sizes.

Fertility and population over time

A temporary fertility increase can leave a permanently larger population. Let Y_t and O_t count young and old households, $\bar{n}_t$ denote mean young fertility, and ν convert children into entering households. Two paths starting from the same young cohort satisfy:

$$\frac{Y_t^1}{Y_t^0} = \prod_{s=0}^{t-1} \frac{\bar{n}_s^1}{\bar{n}_s^0}, \quad O_t^j = Y_{t-1}^j. \quad (6)$$

Population reflects cumulative fertility differences. Both positive steady states have replacement fertility $\bar{n}^* = 1/\nu$. Housing clearing gives:

$$N_{hh}^* = 2Y^* = \frac{2\bar{H}}{\bar{h}^{Y^*} + \bar{h}^{O^*}}. \quad (7)$$

With a fixed stock, a larger population requires less housing per household on average. This stationary ranking also applies to persons under unchanged household and child-counting conventions.

The model admits local equilibrium transitions in which a small permanent increase in the financed share raises initial fertility and final population. The result is established near an economy with all owners, zero child goods cost, and zero property tax. It permits heterogeneous entrant income and wealth and small positive renter shares and costs, provided the same household constraints continue to bind. The purchase limit remains restrictive, saving is positive, and physical owner, old retention, and estate bounds are slack. Convergence and the initial fertility sign require separate parameter conditions.¹

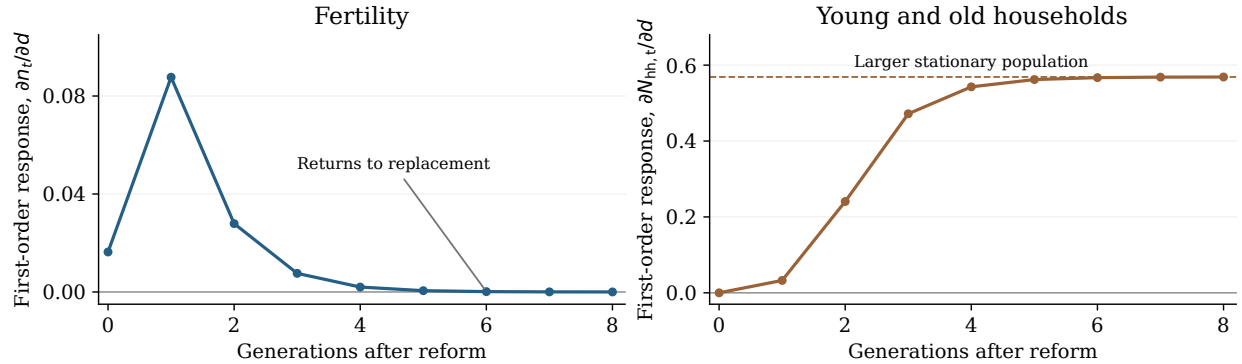


Figure 2. Analytical first-order equilibrium response in the illustrative all-owner limit, per unit increase in $d = b/(1 - \phi)$. Fertility rises most in the second young cohort and returns to replacement; population converges to a higher level. All-date signs hold for small finite reforms in this example; the nearby mixed-tenure result establishes initial fertility and final population signs.

In this limit, the stationary price rises in proportion to borrowing capacity. Each young household's housing and fertility return to their original levels. Older households use less housing, and the stock accommodates more households. Stationary entrants of each type have lower lifetime utility. This transition illustrates a demographic response to credit; the compensated welfare gain in Proposition 1 is a separate comparison.

¹The conditions and proof are in Appendix B of the accompanying longer assessment and sections 5–10 of its supporting proof. The permitted reform size and positive-parameter neighborhood are local and have no numerical bound.

For discussion: how much theory to retain

The allocation result can carry the main claim. It identifies a feasible welfare gain, explains why housing can remain with the old, and requires one short proof. The planner’s borrowing permission is the one already accepted. A complete first-best allocation and a market-based implementation remain separate questions.

The transition proof is useful, but its supporting assumptions are substantive. The stability inequality itself is fairly simple. It does not remove the following restrictions:

Assumption	Economic consequence
Near-all ownership and small child goods costs	The proved neighborhood gives limited scope for tenure switching and the goods cost of children. Neither its width nor a large renter share is covered.
Old owners can sell; estate bounds are slack	The interior formula requires slack retention and, at zero property tax, $(1 - q)\omega_B > q\gamma$. A binding bound changes old housing demand.
Fixed entrant endowments and world bond price	Larger cohorts receive the same distribution of income and wealth. Wages, interest rates, and inherited wealth do not adjust to the larger population.

A small reform is a reasonable scope for an illustrative result. Uniformly unchanged household constraints keep the local proof manageable. The stronger limitations are the economic starting point and the fixed income and wealth of new entrants. This is an existence result for a class of economies, with explicit conditions; its assumptions have not been disciplined by data.

The welfare loss is not peculiar to the plotted parameters. At the all-owner limit with $\chi = \tau^p = 0$, let $d_i = b_i/(1 - \phi)$ and hold the entrant distribution fixed. For each type, the stationary owner value satisfies:

$$\frac{dW^{O*}(i)}{d\phi} = -\frac{(1 - q)d_i/x_i^* + \beta\gamma}{1 - \phi} < 0. \quad (8)$$

Higher prices use more of the household’s resources without increasing its stationary young housing. Old-age housing also falls. The formula holds on the stated owner branch with fixed preferences; ownership tastes are held fixed in the comparison. It compares the same type in two steady states, without assigning a social value to the change in population or evaluating the transitional cohorts.

My recommendation. Use the allocation result and the conditional fertility condition as the core. Keep both figures available, presenting the transition as a separate illustrative extension and putting its proof in the appendix. Its value is the connection between temporary fertility and lasting population, with the price response made explicit. Whether that extension earns main-text space is the next useful discussion. Constrained efficiency and public lending can remain later branches. No model convention or planner power is changed by this proposed presentation.